

Electroweak couplings fixed by Kaluza–Klein moduli: a parameter-free weak mixing angle and fine-structure constant

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The weak mixing angle and the fine-structure constant are independent inputs of the Standard Model. We argue they are fixed by the geometry of electroweak symmetry breaking. Kaluza–Klein reduction makes a gauge coupling a function of the compactification size but leaves that size a free modulus, so the coupling is geometric yet undetermined. We propose that the order parameter of electroweak breaking is the field that stabilizes the modulus: one act of breaking gives the gauge bosons their mass and fixes the size, and with it the couplings. Fermion chirality forces the construction to the ten-dimensional point of an interpolation between an unbroken eleven-dimensional endpoint and a broken nine-dimensional one, evading the chirality obstruction of eleven-dimensional supergravity. A breaking operator at each internal level realizes this: its upper branch is the gauge sector, fixing $\sin^2 \theta$ as a ratio; its lower branch is the order parameter, which—given the assignments stated below—fixes α as a pure number, $1/\alpha = 135.3$, in the infrared. The parameter-free numerical agreement is the evidence.

The claim.—The electroweak sector carries two dimensionless parameters, the weak mixing angle $\sin^2 \theta$ and the fine-structure constant α , taken as measured inputs. We argue that the geometry of symmetry breaking fixes both, through a single mechanism: the field that breaks the electroweak symmetry is the field that fixes the size of the compact dimensions. The argument runs in five steps, and the numbers enter only at the end, as its test.

Kaluza–Klein makes a coupling geometric, then leaves it free.—In a Kaluza–Klein theory the four-dimensional gauge couplings are determined by the internal geometry. Witten’s analysis states it plainly: the couplings scale as a power of $1/(M_p R)$, with R the radius of the extra dimensions, so the coupling *is* the size [1]. The same analysis states the obstruction: “the field equations do not determine the radius of the circle,” which appears as “a massless scalar degree of freedom” [1]. The size is a modulus. A Kaluza–Klein theory therefore predicts a coupling only once something fixes that modulus. Identifying what fixes it is the content of this Letter.

Chirality points to ten dimensions.—The unbroken gauge group $SU(3) \times SU(2) \times U(1)$ requires seven internal dimensions, hence total $D = 11$; but Witten proved that this seven-dimensional compactification cannot produce chiral fermions with the correct quantum numbers [1]. The physical world is therefore not at that endpoint. Electroweak breaking moves the system along the interpolation that ends, when the breaking is complete, at $SU(3) \times U(1)_{\text{em}}$ in five internal dimensions ($D = 9$). Its interior point sits at six internal dimensions, $D = 10$ —the critical dimension of the superstring, where chiral fermions are standard. Witten’s no-go constrains the $D = 11$ endpoint and leaves a $D = 10$ interior unconstrained. We take this as motivation only: the interpolation is heuristic, with no interpolating field exhibited through a noninteger dimension, yet it places the construction at $D = 10$, where the string (Regge) excitations

and the relevant moduli also live.

Breaking is moduli stabilization.—At each internal level, labelled by the Laplacian eigenvalue $J = s(s+1)$, a gauge mode and an order-parameter mode mix into one operator,

$$Q(J) = \mu^2 \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad x_{\pm}(J) = \frac{1}{2} \left(-J \pm \sqrt{J^2 + 4J} \right), \quad (1)$$

with eigenvalues $\mu^2 x_{\pm}$ and the invariant $x_+ x_- = -J$, equivalently $M_+^2 M_-^2 = -\mu^4 J$: one operator sets both branches together. The family $x^2 + Ax - A = 0$ has golden roots at $A = 1$, and (1) is its angular-momentum member $A = s(s+1)$, so the golden ratio is the special level $s = 1/\varphi$. The off-diagonal $\sqrt{J}$ is geometric: the exact one-form $d\phi$ built from a level- J scalar satisfies $\|d\phi\|^2 = J\|\phi\|^2$, so the gauge–scalar mixing is $\sqrt{J}$. This off-diagonal is the supercharge of an $N = 2$ supersymmetric quantum mechanics: with the exterior derivative d and its adjoint δ as supercharges, $\{d, \delta\} = \Delta$ is the Hodge Laplacian, pairing the scalar and vector modes degenerately at eigenvalue J [2]; the diagonal $-J$ breaks this supersymmetry and lifts the degeneracy, so the two branches are the split would-be superpartners—a broken $N = 2$ supersymmetric quantum mechanics, the operator content of the original “susy-like” degeneration [3]. The upper branch $x_+ > 0$ is the massive gauge mode; the lower branch $x_- < 0$ is tachyonic. A negative mass squared is not a pathology on a compact internal space: such a mode is admissible when it respects the Breitenlohner–Freedman bound, and a stabilized negative mode is what selects a new vacuum radius, as in the squashed-sphere deformations of Kaluza–Klein supergravity [4, 5]. Its condensation *is* the symmetry breaking and the stabilization of the modulus: the lower branch is the order parameter that fixes R . That the electroweak sector and the compactification modulus are

tied in this way is established in extra-dimensional models, where the Higgs and the radion mix or the Higgs itself stabilizes the radius [6, 7], and in gauge-Higgs unification, where the order parameter is a component of the higher-dimensional gauge field with its potential fixed by the compactification [8, 9]. We adopt, as a hypothesis, the strongest form of this link, in which the order parameter *is* the stabilizer. One operator then does both jobs—it gives the gauge bosons mass and fixes the size that Witten left free.

Hence the couplings.—Assign the upper branch to the massive vectors, $M_V^2 = \mu^2 x_+(J_V)$, with the charged sector at the doublet value $J = \frac{3}{4}$ and the neutral sector at the adjoint value $J = 2$ (the label s in $J = s(s+1)$ is an internal representation index, distinct from the Lorentz spin and the Regge level). The mixing angle is then a ratio within this one branch,

$$\sin^2 \theta = 1 - \frac{M_W^2}{M_Z^2} = 1 - \frac{x_+(3/4)}{x_+(2)} = 0.223101. \quad (2)$$

Once the lower branch fixes R , the gauge coupling is no longer free. Because the gauge mass M_W and the order parameter $v = \sqrt{2} M_-(2)$ are eigenvalues of the *same* operator, both proportional to μ , the scale cancels:

$$g^2 = \frac{4M_W^2}{v^2} = x_+(3/4) x_+(2),$$

$$\frac{1}{\alpha} = \frac{4\pi}{x_+(3/4) [x_+(2) - x_+(3/4)]} = 135.3. \quad (3)$$

The coupling is read where the modulus is fixed—the compact, infrared scale—so it should match the running coupling there, well below M_Z . The measured $1/\alpha$ runs from 137.036 at zero momentum to 127.955 at M_Z [10]; the value 135.3 sits at the low-energy end, in the few-GeV/hadronic region, where the comparison carries the hadronic vacuum-polarization uncertainty [11]. The picture places α in the infrared, far from M_Z ; the precise scale awaits that uncertainty.

One scale, two dual branches.—The defining feature of $Q(J)$ is that its dimensionless eigenvalues have equal trace and determinant, $x_+ + x_- = x_+ x_- = -J$. Their reciprocals therefore partition unity, $1/x_+ + 1/x_- = 1$, that is

$$\frac{1}{M_+^2} + \frac{1}{M_-^2} = \frac{1}{\mu^2}, \quad \text{tr } Q(J)^{-1} = \frac{1}{\mu^2}, \quad (4)$$

for every level: the inverse operator carries a level-independent trace set by the one scale. Each gauge mass combines in parallel with its order-parameter partner to μ^2 ; the Z slot gives M_Z^2 as the parallel combination of μ^2 and $M_-^2(2)$. The two branches are exchanged by the inversion $x \rightarrow -J/x$, equivalently $M^2 \rightarrow -\mu^4 J/M^2$, an ultraviolet-infrared duality that maps the light gauge branch onto the heavy order-parameter branch and is

TABLE I. What the construction derives versus what it assigns. The derived rows are exact algebra; the assigned rows are the Kaluza–Klein identifications a reduced-action derivation must supply.

Derived (algebra)	Assigned (KK identification)
$x^2 + Jx - J = 0$, roots $x_{\pm}$	operator $Q(J)$ from a $D=10$ reduction
$x_+ x_- = x_+ + x_- = -J$	sign of the $-J$ entry (negative mode)
$1/M_+^2 + 1/M_-^2 = 1/\mu^2$	upper branch $\leftrightarrow$ gauge bosons
inversion $x \rightarrow -J/x$	$(J_W, J_Z) = (3/4, 2)$
$g^2 = x_+(3/4) x_+(2)$, $\sin^2 \theta$, α $v = \sqrt{2} M_-(2)$	

TABLE II. Both branches from one input, $\mu = 106.578$ GeV fixed by M_Z . Reference values from [10].

	level	branch [GeV]	reference [GeV]
W (gauge)	$x_+(3/4)$	80.375	80.369
Z (gauge)	$x_+(2)$	91.188	91.188 (input)
order param. (h?)	$x_-(3/4)$	122.39	125.20
order param. (t?)	$x_-(2)$	176.16	172.57
vacuum v	–	249.13	246.22

fixed by the same trace–determinant condition. The construction is in this sense self-dual at each level, the gauge and order-parameter sectors being two faces of one operator. The large-level behaviour identifies those faces against the spin s : $M_+ \rightarrow \mu$ is bounded (a point, the $p=0$ rotating-brane law $M \sim \text{const}$), while $M_- \propto s$ (since $|x_-| \rightarrow J = s(s+1)$), the $p \rightarrow \infty$ space-filling case. The inversion exchanges a point-like (D0) and a space-filling branch, an all-Dirichlet/all-Neumann T-dual pair.

Evidence.—The claim is the mechanism; the numbers test it. First, (2) equals the on-shell value $1 - M_W^2/M_Z^2 = 0.22320$ from $M_W = 80.369$, $M_Z = 91.188$ GeV [10] to 0.04%; it is the pole-mass angle, distinct from the effective (0.23155) and $\overline{\text{MS}}$ (0.23129) definitions [12]. Second, fixing the one scale by $M_Z^2 = \mu^2 x_+(2)$ gives $\mu = 106.578$ GeV and the spectrum of Table II: the upper branch returns W , the lower branch returns scalar and fermion scales near the Higgs and top and the vacuum, each within a percent—from one input. Third, the upper branch continues to higher levels: $s = 3/2$ places a state at $M_+ = \mu \sqrt{x_+(15/4)} = 96.5$ GeV. The upper branch carries bosons, so this is a further neutral boson; were it the reported ~ 95 GeV diphoton feature [13] it would be neutral and spin-0, which fixes its quantum numbers. We assign it no electric charge: a single chiral charged state would not complete a Dirac fermion under the vector-like electromagnetic $U(1)$, and a $\gamma\gamma$ resonance is in any case neutral.

Discussion.—The result is the mechanism: electroweak symmetry breaking is Kaluza–Klein moduli stabilization, so the order parameter that breaks the symmetry fixes the compactification and, with it, the two dimensionless

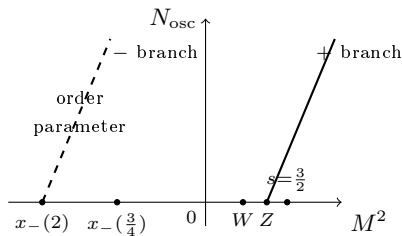


FIG. 1. Schematic Chew–Frautschi reading: the two branches are the $N_{\text{osc}} = 0$ intercepts of common-slope trajectories $M_{N_{\text{osc}},j,\pm}^2 = \mu^2 x_{\pm}(j(j+1)) + N_{\text{osc}}/\alpha'$. Positive intercepts carry the gauge bosons (W , Z) and higher neutral states; negative (tachyonic) intercepts carry the order-parameter scales. Not to scale.

couplings; chirality places the construction in ten dimensions; and α is an infrared quantity because it is read where the modulus is fixed. The numbers are the evidence that the mechanism is real.

Two derivations remain, and we state them as such. The internal six-manifold whose mode reduction yields (1) is not identified here: the geometric mixing $\sqrt{J}$ and the massless gauge entry follow from the Hodge structure, while the tachyonic entry $-J$ requires the internal curvature to supply the order-parameter mass at each level. Equal trace and determinant force (off-diagonal)² = $-(\text{lower entry})$, so $\sqrt{J}$ and $-J$ are one internal datum, the Laplacian eigenvalue and its root; the open input is the single sign that makes the order-parameter mode tachyonic, a Breitenlohner–Freedman-admissible negative mode on a squashed internal space such as $S^3 = SU(2)$, whose Casimir is $j(j+1)$. The half-integer slot $J = 3/4$ ($s = \frac{1}{2}$) requires the spinor harmonics of $S^3 = SU(2)$, absent on the coset S^2 , which carries only integer l . And the assignment $(J_W, J_Z) = (\frac{3}{4}, 2)$ remains to be derived from the gauge-Higgs reduction. Each level extends to a common-slope Regge trajectory $M_{N_{\text{osc}},j,\pm}^2 = \mu^2 x_{\pm}(j(j+1)) + N_{\text{osc}}/\alpha'$, with the sector label j , the oscillator level N_{osc} , and the physical spin kept distinct and the present states at $N_{\text{osc}} = 0$ (Fig. 1)—the

string excitations of the same ten-dimensional compactification. This structure is developed in a companion long version. The claim of this Letter stands on its own: the geometry that fixes the extra dimensions fixes the electroweak couplings.

The construction originates in the de Vries–Rivero observation [3].

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